Aurora: Module Relay x8

Stakeholders Requirements

Prepared for

Engineering Team

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Disclosure is prohibited

Revision History

Version	Date	Author	Description
0.1.0	2026-07-02	Martin Pellizzer	Starting version

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Revision History 2

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Stakeholder 154

requirements

Requirement 1

Requirement

The Relay Module shall execute relay-output commands received from CORE

Answer

The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.

Question

What exactly is the responsibility of the Relay Module within Aurora?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	1
Answer ID	1
Stakeholder Role	System Architect
Question Text	What exactly is the responsibility of the Relay Module within Aurora?
Answer Text	The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.
Requirement Text	The Relay Module shall execute relay-output commands received from CORE

Requirement 2

Requirement

The Relay Module shall provide CORE with output status

Answer

The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.

Question

What exactly is the responsibility of the Relay Module within Aurora?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	1
Answer ID	1
Stakeholder Role	System Architect
Question Text	What exactly is the responsibility of the Relay Module within Aurora?
Answer Text	The Relay Module shall reliably and deterministically execute relay-output commands received from the CORE, provide the CORE with output status and module diagnostics, and transition its outputs to defined safe states when required by system-level fault conditions.
Requirement Text	The Relay Module shall provide CORE with output status

Requirement 3

Requirement

The Relay Module shall provide CORE with module diagnostics

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall provide CORE with module diagnostics

Requirement 4

Requirement

The Relay Module shall transition its outputs to safe states when system-level fault

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall transition its outputs to safe states when system-level fault

Requirement 5

Requirement

The Relay Module shall provide eight indipendently controlled relay-driven channels

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall provide eight indipendently controlled relay-driven channels

Requirement 6

Requirement

The Relay Module shall manage activation of each channel

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall manage activation of each channel

Requirement 7

Requirement

The Relay Module shall manage deactivation of each channel

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall manage deactivation of each channel

Requirement 8

Requirement

The Relay Module shall monitor output faults

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall monitor output faults

Requirement 9

Requirement

The Relay Module shall communicate channels status to CORE

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall communicate channels status to CORE

Requirement 10

Requirement

The Relay Module shall communicate channels diagnostics to CORE

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall communicate channels diagnostics to CORE

Requirement 11

Requirement

The Relay Module shall communicate to CORE via Modbus RTU protocol

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall communicate to CORE via Modbus RTU protocol

Requirement 12

Requirement

The Relay Module shall communicate to CORE via RS-485 hardware standard

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
Answer ID	2
Stakeholder Role	System Architect
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?
Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall communicate to CORE via RS-485 hardware standard

Requirement 13

Requirement

The Relay Module shall enforce local fault-handling

Answer

The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.

Question

Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	2
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Answer Text	The Relay Module shall provide eight independently controlled relay-driver channels, manage their activation and deactivation, monitor relevant module/output faults, communicate their status and diagnostics to the CORE via Modbus RTU over RS-485, and enforce defined local fault-handling behavior.
Requirement Text	The Relay Module shall enforce local fault-handling

Requirement 14

Requirement

The Relay Module shall turn ON/OFF outputs within specified time

Answer

Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.

Question

What behavior does the system require from each of the eight relay outputs?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	3
Answer ID	3
Stakeholder Role	System Architect
Question Text	What behavior does the system require from each of the eight relay outputs?
Answer Text	Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.
Requirement Text	The Relay Module shall turn ON/OFF outputs within specified time

Requirement 15

Requirement

The Relay Module shall prevent one faulty channel to affect other channels

Answer

Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.

Question

What behavior does the system require from each of the eight relay outputs?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	3
Answer ID	3
Stakeholder Role	System Architect
Question Text	What behavior does the system require from each of the eight relay outputs?
Answer Text	Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.
Requirement Text	The Relay Module shall prevent one faulty channel to affect other channels

Requirement 16

Requirement

The Relay Module shall maintain the output defined safe behavior under fault conditions

Answer

Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.

Question

What behavior does the system require from each of the eight relay outputs?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	3
Answer ID	3
Stakeholder Role	System Architect
Question Text	What behavior does the system require from each of the eight relay outputs?
Answer Text	Each output shall independently assume the ON or OFF state commanded by the CORE within the specified response time, while preventing an individual channel fault from affecting the other channels and maintaining the output's defined safe behavior under fault conditions.
Requirement Text	The Relay Module shall maintain the output defined safe behavior under fault conditions

Requirement 17

Requirement

The Relay Module shall detect communication timeout with CORE

Answer

Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.

Question

What must happen to every relay output when communication with the CORE is lost?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	4
Answer ID	4
Stakeholder Role	System Architect
Question Text	What must happen to every relay output when communication with the CORE is lost?
Answer Text	Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.
Requirement Text	The Relay Module shall detect communication timeout with CORE

Requirement 18

Requirement

The Relay Module shall transition to its outputs to predefined safe state when timeout with CORE

Answer

Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.

Question

What must happen to every relay output when communication with the CORE is lost?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	4
Answer ID	4
Stakeholder Role	System Architect
Question Text	What must happen to every relay output when communication with the CORE is lost?
Answer Text	Upon detecting a communication timeout with the CORE, every relay output shall transition to its predefined safe state, which shall be OFF unless a system-level hazard analysis explicitly requires another state.
Requirement Text	The Relay Module shall transition to its outputs to predefined safe state when timeout with CORE

Requirement 19

Requirement

The Relay Module shall transition to defined safe state when in failure state

Answer

Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.

Question

What is the required system behavior if the Relay Module itself fails?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	5
Answer ID	5
Stakeholder Role	System Architect
Question Text	What is the required system behavior if the Relay Module itself fails?
Answer Text	Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.
Requirement Text	The Relay Module shall transition to defined safe state when in failure state

Requirement 20

Requirement

The Relay Module shall prevent unintended activation when in failure state

Answer

Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.

Question

What is the required system behavior if the Relay Module itself fails?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	5
Answer ID	5
Stakeholder Role	System Architect
Question Text	What is the required system behavior if the Relay Module itself fails?
Answer Text	Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.
Requirement Text	The Relay Module shall prevent unintended activation when in failure state

Requirement 21

Requirement

The Relay Module shall provide sufficient fault information to CORE when in failure state

Answer

Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.

Question

What is the required system behavior if the Relay Module itself fails?

Stakeholder

System Architect

Field	Value
Stakeholder ID	1
Question ID	5
Answer ID	5
Stakeholder Role	System Architect
Question Text	What is the required system behavior if the Relay Module itself fails?
Answer Text	Any Relay Module failure that could compromise safe operation shall cause the affected outputs to transition to their defined safe states, prevent unintended activation, and provide sufficient fault information to the CORE to initiate the appropriate Aurora system fault response.
Requirement Text	The Relay Module shall provide sufficient fault information to CORE when in failure state

Requirement 22

Requirement

The Relay Module shall execute commands deterministically

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Requirement Text	The Relay Module shall execute commands deterministically

Requirement 23

Requirement

The Relay Module shall manage its eight physical driver outputs

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Requirement Text	The Relay Module shall manage its eight physical driver outputs

Requirement 24

Requirement

The Relay Module shall perform local hardware diagnostics

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Requirement Text	The Relay Module shall perform local hardware diagnostics

Requirement 25

Requirement

The Relay Module shall report its status to CORE

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Requirement Text	The Relay Module shall report its status to CORE

Requirement 26

Requirement

The Relay Module shall report its faults to CORE

Answer

The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.

Question

Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	6
Answer ID	6
Stakeholder Role	CORE Firmware Engineer
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?
Answer Text	The CORE shall own all system-level decisions, sequencing, interlocks, operating modes and desired relay states, while the Relay Module shall execute those commands deterministically, manage its eight physical driver outputs, perform local hardware diagnostics, and report its status and faults to the CORE.
Requirement Text	The Relay Module shall report its faults to CORE

Requirement 27

Requirement

The Relay Module shall report the state of all eight outputs

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report the state of all eight outputs

Requirement 28

Requirement

The Relay Module shall report the module health

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report the module health

Requirement 29

Requirement

The Relay Module shall report individual channel faults

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report individual channel faults

Requirement 30

Requirement

The Relay Module shall report communication status

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report communication status

Requirement 31

Requirement

The Relay Module shall report electrical identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report electrical identification

Requirement 32

Requirement

The Relay Module shall report thermal identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report thermal identification

Requirement 33

Requirement

The Relay Module shall report diagnostics identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report diagnostics identification

Requirement 34

Requirement

The Relay Module shall report firmware identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report firmware identification

Requirement 35

Requirement

The Relay Module shall report hardware identification

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report hardware identification

Requirement 36

Requirement

The Relay Module shall report any condition preventing reliable output control

Answer

The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.

Question

What information must the Relay Module provide back to the CORE?

Stakeholder

CORE Firmware Engineer

Field	Value
Stakeholder ID	2
Question ID	8
Answer ID	8
Stakeholder Role	CORE Firmware Engineer
Question Text	What information must the Relay Module provide back to the CORE?
Answer Text	The Relay Module shall report the state of all eight outputs, module health, individual channel faults, communication status, relevant electrical/thermal diagnostics, firmware/hardware identification, and any condition preventing reliable output control.
Requirement Text	The Relay Module shall report any condition preventing reliable output control

Requirement 37

Requirement

The Relay Module shall reliably energize its assigned relay coil across the full specified supply

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall reliably energize its assigned relay coil across the full specified supply

Requirement 38

Requirement

The Relay Module shall independently energize its assigned relay coil across the full specified supply

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall independently energize its assigned relay coil across the full specified supply

Requirement 39

Requirement

The Relay Module shall energize its assigned relay coil across the full specified temperature

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil across the full specified temperature

Requirement 40

Requirement

The Relay Module shall energize its assigned relay coil across the full specified tolerance range

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil across the full specified tolerance range

Requirement 41

Requirement

The Relay Module shall energize its assigned relay coil with adequate voltage

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil with adequate voltage

Requirement 42

Requirement

The Relay Module shall energize its assigned relay coil with adequate voltage margin

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil with adequate voltage margin

Requirement 43

Requirement

The Relay Module shall energize its assigned relay coil with adequate current margin

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil with adequate current margin

Requirement 44

Requirement

The Relay Module shall energize its assigned relay coil with continuous-duty

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil with continuous-duty

Requirement 45

Requirement

The Relay Module shall energize its assigned relay coil with continuous-duty capability

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil with continuous-duty capability

Requirement 46

Requirement

The Relay Module shall energize its assigned relay coil with short-circuit protection

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil with short-circuit protection

Requirement 47

Requirement

The Relay Module shall energize its assigned relay coil with overload protection

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil with overload protection

Requirement 48

Requirement

The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions

Requirement 49

Requirement

The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions

Answer

Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.

Question

What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	11
Answer ID	11
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?
Answer Text	Each driver shall independently and reliably energize its assigned relay coil across the full specified supply, temperature, and tolerance range, with adequate voltage/current margin, continuous-duty capability, short-circuit/overload protection, and no unintended activation during reset or fault conditions.
Requirement Text	The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions

Requirement 50

Requirement

The Relay Module shall operate with coils that have a defined nominal voltage

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have a defined nominal voltage

Requirement 51

Requirement

The Relay Module shall operate with coils that have allowable voltage range

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have allowable voltage range

Requirement 52

Requirement

The Relay Module shall operate with coils that have nominal current

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have nominal current

Requirement 53

Requirement

The Relay Module shall operate with coils that have maximum current

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have maximum current

Requirement 54

Requirement

The Relay Module shall operate with coils that have defined resistance

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have defined resistance

Requirement 55

Requirement

The Relay Module shall operate with coils that have defined inductance

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have defined inductance

Requirement 56

Requirement

The Relay Module shall operate with coils that have defined inrush behavior

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have defined inrush behavior

Requirement 57

Requirement

The Relay Module shall operate with coils that have defined continuous-duty rating

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have defined continuous-duty rating

Requirement 58

Requirement

The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver

Requirement 59

Requirement

The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver

Answer

Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.

Question

What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	12
Answer ID	12
Stakeholder Role	Electrical Engineer
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?
Answer Text	Each relay coil shall have a defined nominal voltage, allowable voltage range, nominal and maximum current, resistance/inductance, inrush behavior, continuous-duty rating, and release/operate characteristics that are fully compatible with its corresponding driver.
Requirement Text	The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver

Requirement 60

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Requirement Text	The Relay Module shall operate correctly over the complete specified nominal supply range

Requirement 61

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Requirement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances

Requirement 62

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Requirement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation

Requirement 63

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Requirement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions

Requirement 64

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Requirement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts

Requirement 65

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Requirement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage

Requirement 66

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Requirement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

Requirement 67

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet

Answer

The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.

Question

What is the required input supply voltage range, including tolerances and transient conditions?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	13
Answer ID	13
Stakeholder Role	Electrical Engineer
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?
Answer Text	The module shall operate correctly over the complete specified nominal supply range, including component tolerances, steady-state variation, startup conditions, brownouts, overvoltage, ripple, and transient disturbances expected in the Aurora electrical cabinet.
Requirement Text	The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet

Requirement 68

Requirement

The Relay Module shall have drivers with appropriately sized suppressions

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Requirement Text	The Relay Module shall have drivers with appropriately sized suppressions

Requirement 69

Requirement

The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Requirement Text	The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode

Requirement 70

Requirement

The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Requirement Text	The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

Requirement 71

Requirement

The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Requirement Text	The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings

Requirement 72

Requirement

The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays

Answer

Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.

Question

What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	14
Answer ID	14
Stakeholder Role	Electrical Engineer
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?
Answer Text	Each driver output shall incorporate appropriately sized suppression, such as a flyback diode or TVS/clamp network, to limit the coil-generated transient to a safe level without exceeding driver ratings or causing unacceptable relay release-time delays.
Requirement Text	The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays

Requirement 73

Requirement

The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout

Answer

During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.

Question

What electrical behavior is required during power loss, brownout, and restoration of power?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	15
Answer ID	15
Stakeholder Role	Electrical Engineer
Question Text	What electrical behavior is required during power loss, brownout, and restoration of power?
Answer Text	During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.
Requirement Text	The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout

Requirement 74

Requirement

The Relay Module shall transition all relay outputs to defined safe state during power loss

Answer

During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.

Question

What electrical behavior is required during power loss, brownout, and restoration of power?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	15
Answer ID	15
Stakeholder Role	Electrical Engineer
Question Text	What electrical behavior is required during power loss, brownout, and restoration of power?
Answer Text	During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.
Requirement Text	The Relay Module shall transition all relay outputs to defined safe state during power loss

Requirement 75

Requirement

The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

Answer

During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.

Question

What electrical behavior is required during power loss, brownout, and restoration of power?

Stakeholder

Electrical Engineer

Field	Value
Stakeholder ID	3
Question ID	15
Answer ID	15
Stakeholder Role	Electrical Engineer
Question Text	What electrical behavior is required during power loss, brownout, and restoration of power?
Answer Text	During power loss or an out-of-range brownout, all relay outputs shall transition to and remain in their defined safe state without unintended activation, and after power restoration the module shall initialize deterministically with outputs disabled until valid control from the CORE is established.
Requirement Text	The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

Requirement 76

Requirement

The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Requirement Text	The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels

Requirement 77

Requirement

The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Requirement Text	The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels

Requirement 78

Requirement

The Relay Module shall detect relevant hardware faults

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Requirement Text	The Relay Module shall detect relevant hardware faults

Requirement 79

Requirement

The Relay Module shall detect relevant output faults

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Requirement Text	The Relay Module shall detect relevant output faults

Requirement 80

Requirement

The Relay Module shall transition affected outputs to their predefined safe states independently of higher-level system decisions

Answer

The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.

Question

What safety functions, if any, are assigned specifically to the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	16
Answer ID	16
Stakeholder Role	Safety Engineer
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?
Answer Text	The Relay Module shall locally enforce the safety-related output constraints and fail-safe behavior defined for its eight relay channels, detect relevant hardware and output faults, and transition affected outputs to their predefined safe states independently of higher-level system decisions.
Requirement Text	The Relay Module shall transition affected outputs to their predefined safe states independently of higher-level system decisions

Requirement 81

Requirement

The Relay Module shall have a formally defined safe state for all its outputs based on its associated hazard analysis

Answer

Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.

Question

What is the required safe state for each of the eight relay outputs?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	17
Answer ID	17
Stakeholder Role	Safety Engineer
Question Text	What is the required safe state for each of the eight relay outputs?
Answer Text	Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.
Requirement Text	The Relay Module shall have a formally defined safe state for all its outputs based on its associated hazard analysis

Requirement 82

Requirement

The Relay Module shall have OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized

Answer

Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.

Question

What is the required safe state for each of the eight relay outputs?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	17
Answer ID	17
Stakeholder Role	Safety Engineer
Question Text	What is the required safe state for each of the eight relay outputs?
Answer Text	Each of the eight relay outputs shall have a formally defined safe state based on its associated hazard analysis, with OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized.
Requirement Text	The Relay Module shall have OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized

Requirement 83

Requirement

The Relay Module shall immediately prevent unsafe output activation when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Requirement Text	The Relay Module shall immediately prevent unsafe output activation when it detects an internal fault capable of compromising safe operation

Requirement 84

Requirement

The Relay Module shall force all affected outputs to their defined safe states when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Requirement Text	The Relay Module shall force all affected outputs to their defined safe states when it detects an internal fault capable of compromising safe operation

Requirement 85

Requirement

The Relay Module shall latch the fault as required when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Requirement Text	The Relay Module shall latch the fault as required when it detects an internal fault capable of compromising safe operation

Requirement 86

Requirement

The Relay Module shall report the fault as required when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Requirement Text	The Relay Module shall report the fault as required when it detects an internal fault capable of compromising safe operation

Requirement 87

Requirement

The Relay Module shall prevent reactivation until the fault-clear conditions are satisfied when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Requirement Text	The Relay Module shall prevent reactivation until the fault-clear conditions are satisfied when it detects an internal fault capable of compromising safe operation

Requirement 88

Requirement

The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

Answer

When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.

Question

What shall happen to each relay output when the Relay Module detects an internal fault?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	18
Answer ID	18
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?
Answer Text	When the Relay Module detects an internal fault capable of compromising safe operation, it shall immediately prevent unsafe output activation, force all affected outputs to their defined safe states, latch or report the fault as required, and prevent reactivation until the fault-clear and reauthorization conditions are satisfied.
Requirement Text	The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

Requirement 89

Requirement

The Relay Module shall autonomously transition every output to its predefined safe state when communication with the CORE is lost beyond the defined safety timeout

Answer

When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Question

What shall happen to each relay output when communication with the CORE is lost?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	19
Answer ID	19
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when communication with the CORE is lost?
Answer Text	When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.
Requirement Text	The Relay Module shall autonomously transition every output to its predefined safe state when communication with the CORE is lost beyond the defined safety timeout

Requirement 90

Requirement

The Relay Module shall maintain a predefined safe state when communication with the CORE is lost beyond the defined safety timeout

Answer

When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Question

What shall happen to each relay output when communication with the CORE is lost?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	19
Answer ID	19
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when communication with the CORE is lost?
Answer Text	When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.
Requirement Text	The Relay Module shall maintain a predefined safe state when communication with the CORE is lost beyond the defined safety timeout

Requirement 91

Requirement

The Relay Module shall report the communication fault when communication with CORE is restored

Answer

When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Question

What shall happen to each relay output when communication with the CORE is lost?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	19
Answer ID	19
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when communication with the CORE is lost?
Answer Text	When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.
Requirement Text	The Relay Module shall report the communication fault when communication with CORE is restored

Requirement 92

Requirement

The Relay Module shall require explicit reauthorization from CORE before returning to normal operation

Answer

When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.

Question

What shall happen to each relay output when communication with the CORE is lost?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	19
Answer ID	19
Stakeholder Role	Safety Engineer
Question Text	What shall happen to each relay output when communication with the CORE is lost?
Answer Text	When communication with the CORE is lost beyond the defined safety timeout, the Relay Module shall autonomously transition every output to its predefined safe state, maintain that state, report the communication fault when communication is restored, and require explicit reauthorization before returning to normal operation.
Requirement Text	The Relay Module shall require explicit reauthorization from CORE before returning to normal operation

Requirement 93

Requirement

The Relay Module shall locally enforce output-level safety constraints

Answer

The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Question

Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	20
Answer ID	20
Stakeholder Role	Safety Engineer
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?
Answer Text	The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.
Requirement Text	The Relay Module shall locally enforce output-level safety constraints

Requirement 94

Requirement

The Relay Module shall locally enforce watchdog/communication-loss responses

Answer

The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Question

Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	20
Answer ID	20
Stakeholder Role	Safety Engineer
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?
Answer Text	The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.
Requirement Text	The Relay Module shall locally enforce watchdog/communication-loss responses

Requirement 95

Requirement

The Relay Module shall locally enforce internal fault handling

Answer

The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Question

Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	20
Answer ID	20
Stakeholder Role	Safety Engineer
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?
Answer Text	The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.
Requirement Text	The Relay Module shall locally enforce internal fault handling

Requirement 96

Requirement

The Relay Module shall locally enforce defined safe-state behavior without relying solely on the CORE

Answer

The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.

Question

Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Stakeholder

Safety Engineer

Field	Value
Stakeholder ID	4
Question ID	20
Answer ID	20
Stakeholder Role	Safety Engineer
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?
Answer Text	The CORE shall make all system-level safety decisions, operating-mode and shutdown decisions, while the Relay Module shall locally enforce output-level safety constraints, watchdog/communication-loss responses, internal fault handling, and the defined safe-state behavior without relying solely on the CORE.
Requirement Text	The Relay Module shall locally enforce defined safe-state behavior without relying solely on the CORE

Requirement 97

Requirement

The Relay Module shall manage the eight relay outputs to control or influence the ozone-generation process by operating the required process equipment and auxiliaries-such as ozone-generation power, cooling, gas/air or oxygen supply, valves, purge/ventilation, and other process actuators—according to the defined process sequence and interlocks

Answer

The eight relay outputs shall control or influence the ozone-generation process by operating the required process equipment and auxiliaries—such as ozone-generation power, cooling, gas/air or oxygen supply, valves, purge/ventilation, and other process actuators—according to the defined process sequence and interlocks.

Question

What process functions are controlled or influenced by the eight relay outputs of the Relay Module?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	21
Answer ID	21
Stakeholder Role	Process Engineer
Question Text	What process functions are controlled or influenced by the eight relay outputs of the Relay Module?
Answer Text	The eight relay outputs shall control or influence the ozone-generation process by operating the required process equipment and auxiliaries—such as ozone-generation power, cooling, gas/air or oxygen supply, valves, purge/ventilation, and other process actuators—according to the defined process sequence and interlocks.
Requirement Text	The Relay Module shall manage the eight relay outputs to control or influence the ozone-generation process by operating the required process equipment and auxiliaries-such as ozone-generation power, cooling, gas/air or oxygen supply, valves, purge/ventilation, and other process actuators—according to the defined process sequence and interlocks

Requirement 98

Requirement

The Relay Module shall include a startup sequence that verifies all required permissives

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Requirement Text	The Relay Module shall include a startup sequence that verifies all required permissives

Requirement 99

Requirement

The Relay Module shall include a startup sequence that establishes the required gas flow

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Requirement Text	The Relay Module shall include a startup sequence that establishes the required gas flow

Requirement 100

Requirement

The Relay Module shall include a startup sequence that establishes the supporting utilities

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Requirement Text	The Relay Module shall include a startup sequence that establishes the supporting utilities

Requirement 101

Requirement

The Relay Module shall include a startup sequence that enables cooling

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Requirement Text	The Relay Module shall include a startup sequence that enables cooling

Requirement 102

Requirement

The Relay Module shall include a startup sequence that enables other auxiliaries

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Requirement Text	The Relay Module shall include a startup sequence that enables other auxiliaries

Requirement 103

Requirement

The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Requirement Text	The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries

Requirement 104

Requirement

The Relay Module shall include a startup sequence that energizes the ozone-generation system

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Requirement Text	The Relay Module shall include a startup sequence that energizes the ozone-generation system

Requirement 105

Requirement

The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete

Answer

The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.

Question

What is the required sequence of operations for starting ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	22
Answer ID	22
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for starting ozone generation?
Answer Text	The startup sequence shall verify all required permissives, establish the required gas flow and supporting utilities, enable cooling and other auxiliaries, confirm their healthy operating feedback, energize the ozone-generation system, and only then transition Aurora to the normal ozone-production state.
Requirement Text	The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete

Requirement 106

Requirement

The Relay Module shall include a shutdown sequence that disables ozone generation

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Requirement Text	The Relay Module shall include a shutdown sequence that disables ozone generation

Requirement 107

Requirement

The Relay Module shall include a shutdown sequence that maintains the required cooling

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Requirement Text	The Relay Module shall include a shutdown sequence that maintains the required cooling

Requirement 108

Requirement

The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Requirement Text	The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period

Requirement 109

Requirement

The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Requirement Text	The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order

Requirement 110

Requirement

The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

Answer

The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.

Question

What is the required sequence of operations for stopping ozone generation?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	23
Answer ID	23
Stakeholder Role	Process Engineer
Question Text	What is the required sequence of operations for stopping ozone generation?
Answer Text	The shutdown sequence shall first disable ozone generation, maintain the required cooling and gas-flow/purge functions for the specified post-run period, safely de-energize the remaining auxiliaries in the defined order, and finally confirm that the process has reached the stopped state.
Requirement Text	The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

Requirement 111

Requirement

The Relay Module shall permit ozone generation only when all required utilities are available

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
Question Text	What conditions must be satisfied before ozone generation is permitted to start?
Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Requirement Text	The Relay Module shall permit ozone generation only when all required utilities are available

Requirement 112

Requirement

The Relay Module shall permit ozone generation only when gas flow is within their specified limits

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

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Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
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Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Requirement Text	The Relay Module shall permit ozone generation only when gas flow is within their specified limits

Requirement 113

Requirement

The Relay Module shall permit ozone generation only when cooling is within their specified limits

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
Answer ID	24
Stakeholder Role	Process Engineer
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Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Requirement Text	The Relay Module shall permit ozone generation only when cooling is within their specified limits

Requirement 114

Requirement

The Relay Module shall permit ozone generation only when required valves are in their correct states

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

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Stakeholder ID	5
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Stakeholder Role	Process Engineer
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Requirement Text	The Relay Module shall permit ozone generation only when required valves are in their correct states

Requirement 115

Requirement

The Relay Module shall permit ozone generation only when required auxiliaries are in their correct states

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	24
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Stakeholder Role	Process Engineer
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Answer Text	Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.
Requirement Text	The Relay Module shall permit ozone generation only when required auxiliaries are in their correct states

Requirement 116

Requirement

The Relay Module shall permit ozone generation only when no active safety interlock is violated

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

Stakeholder

Process Engineer

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Requirement Text	The Relay Module shall permit ozone generation only when no active safety interlock is violated

Requirement 117

Requirement

The Relay Module shall permit ozone generation only when no active process interlock is violated

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

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Stakeholder

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Requirement Text	The Relay Module shall permit ozone generation only when no active process interlock is violated

Requirement 118

Requirement

The Relay Module shall permit ozone generation only when all required feedback signals are valid

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

Question

What conditions must be satisfied before ozone generation is permitted to start?

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Requirement Text	The Relay Module shall permit ozone generation only when all required feedback signals are valid

Requirement 119

Requirement

The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

Answer

Ozone generation shall be permitted only when all required utilities are available, gas flow and cooling are within their specified limits, required valves and auxiliaries are in their correct states, no active safety or process interlock is violated, all required feedback signals are valid, and the system is in an authorized operating mode.

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Requirement Text	The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

Requirement 120

Requirement

The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

Field	Value
Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
Question Text	What process conditions require ozone generation to stop immediately?
Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Requirement Text	The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected

Requirement 121

Requirement

The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

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Stakeholder

Process Engineer

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Stakeholder ID	5
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Requirement Text	The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow

Requirement 122

Requirement

The Relay Module shall immediately stop ozone generation when detected a loss of required cooling

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

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Process Engineer

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Requirement Text	The Relay Module shall immediately stop ozone generation when detected a loss of required cooling

Requirement 123

Requirement

The Relay Module shall immediately stop ozone generation when detected a critical equipment failure

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

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Question ID	25
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Requirement Text	The Relay Module shall immediately stop ozone generation when detected a critical equipment failure

Requirement 124

Requirement

The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

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Stakeholder ID	5
Question ID	25
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Answer Text	Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.
Requirement Text	The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits

Requirement 125

Requirement

The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

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Stakeholder ID	5
Question ID	25
Answer ID	25
Stakeholder Role	Process Engineer
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Requirement Text	The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives

Requirement 126

Requirement

The Relay Module shall immediately stop ozone generation when detected emergency shutdown

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

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Requirement Text	The Relay Module shall immediately stop ozone generation when detected emergency shutdown

Requirement 127

Requirement

The Relay Module shall immediately stop ozone generation when detected defined critical interlock

Answer

Ozone generation shall stop immediately whenever a condition capable of creating an unsafe or uncontrolled process state is detected, including loss of required gas flow or cooling, critical equipment failure, hazardous process-variable limits, loss of required safety permissives, emergency shutdown, or any other defined critical interlock.

Question

What process conditions require ozone generation to stop immediately?

Stakeholder

Process Engineer

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Requirement Text	The Relay Module shall immediately stop ozone generation when detected defined critical interlock